# KYLE LAU

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## SUMMARY

Full-stack software engineer and founder of Soniphoria, an AI-powered music education platform with approximately 500 active users and up to 6 partner local music schools. Hands-on across the stack: React/TypeScript, Python (FastAPI), Supabase (PostgreSQL), and Cloudflare. Trained in enterprise Java (Jakarta EE, JSF, JPA, EJB) at BCIT; Agile experience from an OpenTree Education internship.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, Java, Ruby, SQL

**Java:** Jakarta EE / Java EE, Jakarta Faces (JSF) + Facelets, CDI, EJB (session and message-driven beans), JPA/JPQL, JDBC, JAX-RS (REST), WildFly, Maven, JUnit 5, REST Assured

**Front End:** React, Vite, Tailwind CSS, Radix UI, TanStack Query, Zod, i18next, Material-UI, Redux, Bootstrap

**Back End & Data:** FastAPI, Node.js/Express, Ruby on Rails, PostgreSQL (Supabase, RLS), MySQL, MongoDB, Redis

**Cloud, DevOps & Testing:** Cloudflare (Workers, R2), Docker, OpenShift, GitHub Actions, Stripe, Vitest, Playwright, pytest, Git, Agile

## EXPERIENCE

**Founder & CEO / Lead Software Engineer | Soniphoria** Mar 2025 – Present

*AI-powered music education platform | Vancouver, BC | soniphoria.app*

- Founded Soniphoria and led development and launch of an AI-assisted practice platform for music students, parents, teachers, and schools; grew to approximately 500 active users.
- Developed the React + TypeScript web app (Vite, Tailwind CSS, Radix UI, TanStack Query, Zod) with role-based workflows for students, parents, teachers, and school admins, covering homework, practice tracking, and progress reports, and a bilingual English/Chinese interface using i18next.
- Built the Python back end: a FastAPI REST service plus an arq/Redis background-job worker, with file storage on Cloudflare R2.
- Designed the Supabase PostgreSQL schema through 95 SQL migrations with Row-Level Security policies for schools, homework, practice, and notifications; integrated Stripe for subscription payments.
- Deployed the web app on Cloudflare Workers with Dockerized API and worker services; set up GitHub Actions CI and a test suite of 350+ Vitest and pytest files.
- Worked with local music schools and teachers to shape features, with up to 6 schools joining the Soniphoria program; designed onboarding flows (invite links, QR-code registration).

**Full-stack Developer Intern | OpenTree Education** Feb 2023 – May 2023

- Built a graded assessment feature with progress tracking and personalized feedback (TypeScript, React, Material-UI, Express, MySQL, Docker).
- Fixed bugs and submitted GitHub pull requests; took part in sprint planning, daily stand-ups, and retrospectives.

## PROJECTS

**Timesheet & Project Management System | BCIT COMP 4911 team project (20+ students)** Jan 2026 – Apr 2026

- Contributed to a web app for timesheets, projects, and work packages: Jakarta EE REST API (JAX-RS, EJB, JPA) on WildFly, MySQL, and an ASP.NET Razor Pages front end, containerized with Docker and deployed to OpenShift.
- Built the work-package module: database tables, role-based access (employees see only their assigned work packages), responsible-engineer assignment, and a rule locking the work-package structure after the first approved charge; added JUnit and REST Assured integration tests.

**Job Opportunities Search Nearby** Feb 2022

- Built a MERN-stack job board with per-user authorization and a Google Maps nearby-job search (React, Redux, Node.js, MongoDB).

**Journey Manager** Aug 2022

- Built a Ruby on Rails app for sharing journeys and comments, with authentication, authorization, likes, Google Maps, and an admin panel (Ruby on Rails, PostgreSQL).

## EDUCATION

**Computer Systems Technology | British Columbia Institute of Technology (BCIT)** Jun 2026

Information Systems option. COMP 3910 (Full-Stack Development for Enterprise Systems): Jakarta Faces, CDI, JDBC, JPA, JAX-RS. COMP 4911 (Developing Enterprise Services): EJB, JPQL, EJB security and transactions, Unified Process.

**CodeCore Developer Bootcamp | CodeCore College, Vancouver** Sep 2022